

1) ANALYSIS OF CONFIGURATION 1

The goal of this first point is to size the reticular structure linking the tanks to the central body of the spacecraft, showed in fig.1, in order to perform a feasible configuration taking into account the *Margin Of Safety* and the structural mass.

DATA

- $m_s = 130 \text{ Kg}$ (tank's mass)
- $n = 5$ (load factor)
- $E_{Al} = 70 \text{ GPa}$ (Young Modulus of aluminum)
- $\sigma_y \text{ Al} = 250 \text{ MPa}$ (Yielding stress of aluminum)
- $\rho_{Al} = 2700 \text{ Kg/m}^3$ (aluminum's density)
- $L = 30 \text{ cm}$

F must be transmitted to the central cylinder through the 3 beams. Each beam is subjected to a reaction force due to the application of F . Using Ritter's section shown in fig.2 it is possible to evaluate the values of the reaction forces. The lower beam is named ① the central beam ② and that on the top ③. The reference frame is shown below in the same figure.

Computing the equilibrium to translation for each direction and moments of forces:

$$\begin{cases} \sum N_x = N_1 \cos 45^\circ + N_2 \sin 45^\circ + N_3 \sin 45^\circ = 0 \\ \sum N_y = -N_1 \sin 45^\circ + N_2 \cos 45^\circ + N_3 \cos 45^\circ - F = 0 \\ \sum M_o = -N_3 \sin 45^\circ * 2L = 0 \end{cases}$$

So it is obtained that:

$N_3 = 0 \text{ N}$ so beam ③ is not loaded;

$N_2 = \frac{F}{\sqrt{2}} = 4508.87 \text{ N}$ so beam ② is in traction;

$N_1 = -N_2 = -4508.87 \text{ N}$ so beam ① is in compression.

Fig.3 shows the load's resultant vectors along each beam and the decomposition of force F as the sum of N_1 and N_2 .

Now it is possible to size the reticular structure considering aluminum material and making hypothesis on the cross section area and shape of the beams that, in this case, have the same length $l = \sqrt{2}L = 42.43 \text{ cm}$.

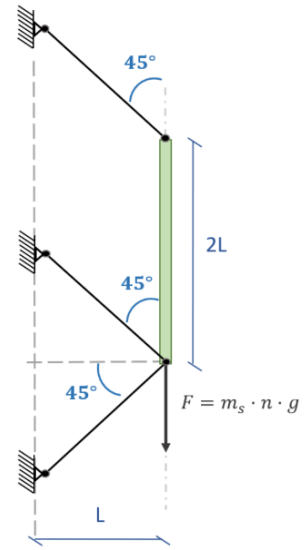


Figure 1:
Configuration 1

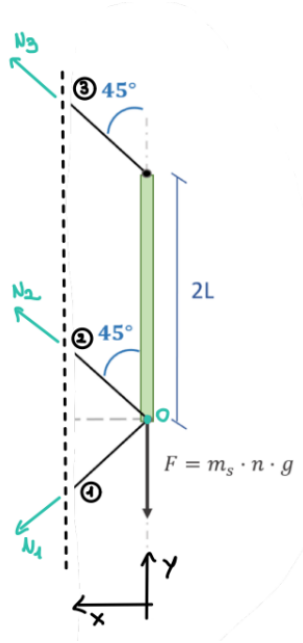


Figure 2: Ritter's section of configuration 1.

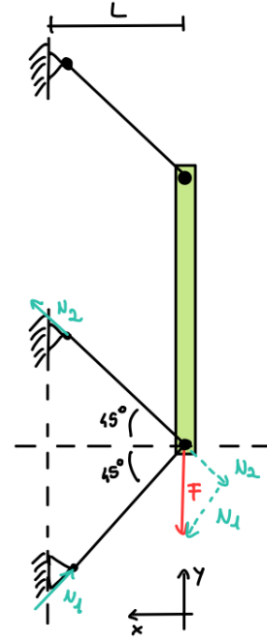


Figure 3: Final load's decomposition for configuration 1.

1.1) MOS EVALUATION

FIRST AREA HP: $A_1 = 1 \text{ cm}^2$

It is possible to evaluate the *Margin Of Safety* due to the admissible load considering that:

$N_{adm} = \sigma_y A_1 = 25000 \text{ N}$ (admissible load);

$FOS = 2$ (given factor of safety);

the project load is given by $N = |N_1| = |N_2|$;

$$MOS_t = \frac{N_{adm}}{FOS * N} - 1 = 1.77 = 177\%$$

which is a very large value so we are oversizing the structure.

Since the beam ① is subjected to a compressive load the previous evaluation of MOS is allowed only for the beam in traction and in this case it is necessary to determine whether it resists to buckling condition. For this configuration, a beam constrained by two hinges, the critical load is $P_{cr} = \frac{\pi^2 E_{AI} I}{l^2}$ where I is the moment of inertia of the considered cross section of the beam.

In this case the admissible load coincides with the critical load and the MOS can be evaluated

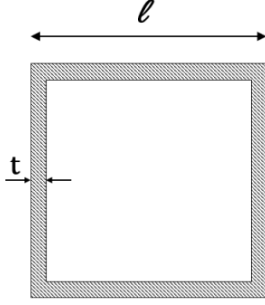


Figure 4: Square thin walled section (1)

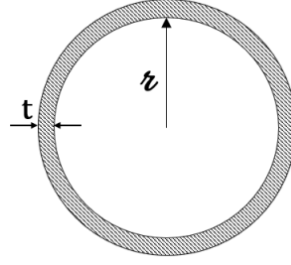


Figure 5: Circular thin walled section (2)

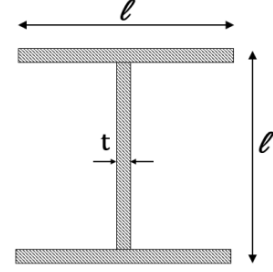


Figure 6: Double T section (3)

as:

$$MOS_c = \frac{P_{cr}}{FOS * N} - 1$$

and, supposing to maintain the previous hypothesis on the area of the cross section, a comparison between different types of cross section (represented in fig.4, fig.5, fig.6) is made obtaining the values of moment of inertia, critical load and MOS showed in tab.1.

	(1)	(2)	(3)
$I [m^4]$	$5.83 * 10^{-9}$	$4.20 * 10^{-9}$	$2.67 * 10^{-9}$
$P_{cr} [N]$	22386	16134	10259
MOS_c	1.48	0.79	0.14

Table 1: Key results for three different cross section area for $A_1 = 1cm^2$

It is worth noting that for the double T section it was considered the moment of inertia around the y axis as the smallest one which can lead to buckling.

Table 1 shows that all the evaluated MOS_c are admissible and very large. It means that it is possible to decrease the cross section area. Computing the solution iterating the process for different cross section areas the best result in terms of MOS_c is obtained for:

BEST AREA HP: $A_2 = 0.65 cm^2$ and the results are shown in table 2.

It can be seen that the best options for the beam in compression are the square thin walled section and the circular thin walled section. Even if the best MOS_c is that of the first section's type, the circular thin walled section has an admissible MOS_c of 24% and from the structural point of view it is the best because it has no preferential directions for the buckling loads, thanks to its symmetrical shape, and it is easier to build than a square section beam.

	(1)	(2)	(3)
$I [m^4]$	$3.98 * 10^{-9}$	$2.92 * 10^{-9}$	$1.74 * 10^{-9}$
$P_{cr} [N]$	15278	11182	6658.7
MOS_c	0.69	0.24	-0.26

Table 2: Key results for three different cross section area for $A_2 = 0.65cm^2$

Since it has been changed the cross section area it is necessary to verify that also the beam in traction has an acceptable MOS_t :

$$N_{adm} = \sigma_y A_2 = 16250 \text{ N}$$

$$MOS_t = 0.80 = 80\% \quad \text{which is an admissible value.}$$

In fig.7 there are the evolutions of both MOS for the beam ① and ② as a function of the circular cross section area.

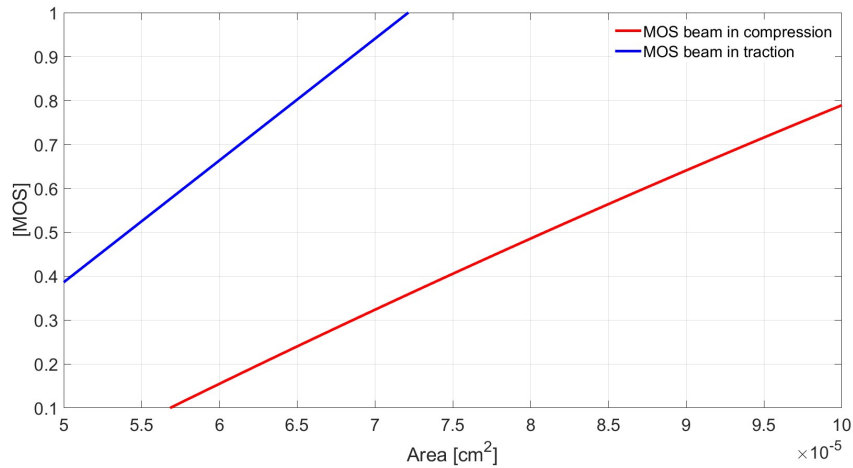


Figure 7: MOS_c and MOS_t trend as a function of the circular cross section area.

1.2) STRUCTURAL MASS EVALUATION

Despite the fact that beam ③ is not loaded it must be taken into account for the evaluation of the structural mass without making any type of load's consideration. It is possible to use the same cross section area of the others two beams.

The structural mass is the sum of the three masses of the beams:

$$M_{str} = 3\rho_{Al}A_2l = 0.2234 \text{ Kg}$$

but since the overall aim is to minimize the structural mass it can be considered the lower possible cross section area for the beam which is not loaded.

2) COMPARISON WITH OTHER CONFIGURATIONS

2.1) ANALYSIS OF CONFIGURATION 2

As in the previous case, by Ritter's section (in fig.9) there can be evaluated the reaction's forces on the three beams. In this case the length of beam ② and the angle between the beam and the tank are:

$$l_2 = \sqrt{(2L)^2 + L^2} = \sqrt{5L^2} = \sqrt{5}L = 67.08 \text{ cm}$$

$$l_2 * \cos\alpha = 2L$$

$$\alpha = \arccos\left(\frac{2L}{l_2}\right) = \arccos\left(\frac{2}{\sqrt{5}}\right) = 26.57^\circ$$

Computing the equilibrium to translation for each direction and moments of forces:

$$\begin{cases} \sum N_x = N_1 + N_3 + N_2 \sin\alpha = 0 \\ \sum N_y = -N_2 \cos\alpha - F = 0 \\ \sum M_o = -N_3 * 2L - N_2 \sin\alpha * 2L = 0 \end{cases}$$

So it is obtained that:

$$N_1 = 0 \text{ N} \quad \text{so beam ① is not loaded;}$$

$$N_2 = -\frac{F}{\cos\alpha} = -7129.45 \text{ N} \quad \text{so beam ② is in compression;}$$

$$N_3 = F * \tan\alpha = 3188.94 \text{ N} \quad \text{so beam ③ is in traction.}$$

In fig.10 there are the final load's decomposition along each beam and the decomposition of force F as the sum of N_2 and N_3 .

Considering the same section of the previous case $A_2 = 0.65 \text{ cm}^2$ the beam ③ is in traction so there is no buckling problem and the MOS_t can be evaluated considering that $N_{adm} = \sigma_y A_2 = 16250 \text{ N}$:

$$MOS_t = \frac{N_{adm}}{FOS * N} - 1 = 1.55$$

Beam ② is in compression so it is necessary to evaluate the critical load. For simplicity, as it was seen that the best section is the circular thin walled section, the calculations will be done assuming $I = 2.92 * 10^{-9} \text{ m}^4$ which is the moment of inertia of this section as shown in tab.2.

$$P_{cr} = \frac{\pi^2 E_{Al} I}{l_2^2} = 4478.72 \text{ N}$$

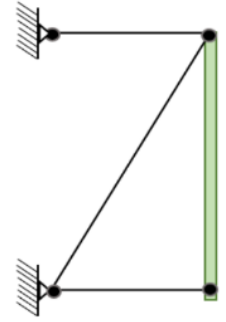


Figure 8:
Configuration 2

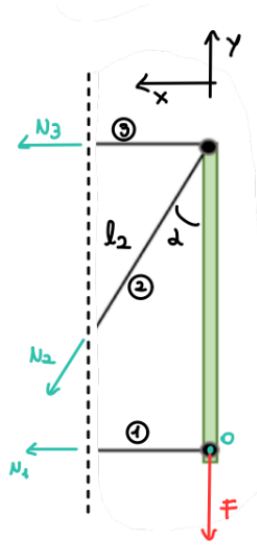


Figure 9: Ritter's section of configuration 2.

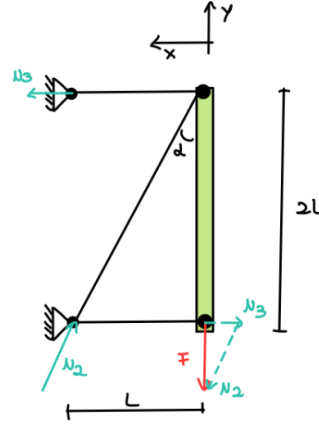


Figure 10: Final load's decomposition for configuration 2.

$$MOS_c = \frac{P_{cr}}{FOS * N_2} - 1 = -0.686$$

In this case beam ② has a margin of safety less than zero, due to the fact that with the considered cross section area the critical load is less than the applied load multiplied by the factor of safety. In other words the beam is not able to resist to buckling condition. This is due to the fact that the beam subjected to the compression load is larger than the one in the first configuration and the best cross section area does not fix with the critical load for the buckling condition. In this case we should consider a larger beam's cross section area at least for beam ②.

2.2) ANALYSIS OF CONFIGURATION 3

Following the same procedure of the previous cases it can be evaluated the force and momentum equilibrium:

$$\begin{cases} \sum N_x = N_1 + N_3 + N_2 \sin \alpha = 0 \\ \sum N_y = N_2 \cos \alpha - F = 0 \\ \sum M_o = -N_3 * 2L = 0 \end{cases}$$

So it is obtained that:

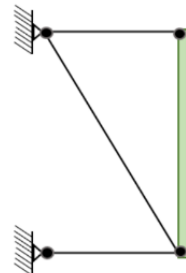


Figure 11:
Configuration 3⁶

$N_3 = 0 \text{ N}$ so beam ③ is not loaded;
 $N_2 = \frac{F}{\cos\alpha} = 7129.45 \text{ N}$ so beam ② is in traction;
 $N_1 = -F * \tan\alpha = -3188.94 \text{ N}$ so beam ① is in compression.

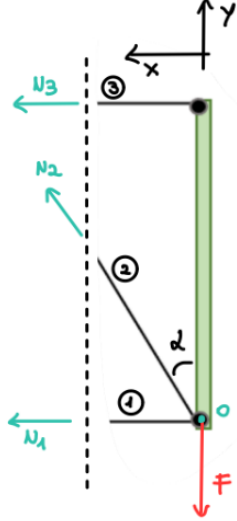


Figure 12: Ritter's section of configuration 3

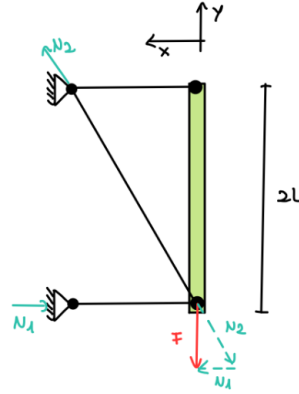


Figure 13: Final load's decomposition for configuration 3

The results are the same as the configuration 2 but with inverse sign for N_2 and inverse values for N_3 and N_1 . Fig.13 shows the vector reaction's loads along each beam and the decomposition of F as the sum of N_1 and N_2 .

The beam ② is in traction and the MOS_t can be evaluated as:

$$MOS_t = \frac{N_{adm}}{FOS * N_2} - 1 = 0.14$$

the beam ① is in compression and the MOS_c is, considering a circular cross section:

$$MOS_c = \frac{P_{cr}}{FOS * |N_1|} - 1 = 2.39 \quad \text{where} \quad P_{cr} = \frac{\pi^2 EI}{l_1^2} = 21625.39 \text{ N}$$

In this case all the MOS are admissible and the structural mass can be evaluated as the sum of the three masses of the beams:

$$M_{str} = \rho A_2 (2l_1 + l_2) = 0.223 \text{ Kg}$$

2.3) RESULTS

Comparing all the 3 configurations (2 excluding the second one which is not admissible from the safety point of view) the best configuration in terms of structural mass is the last one because with the same material and cross section shape and area the structural mass is less than the first configuration (even if the difference is minimal). Therefore in the configuration 3 the beam subjected to compressive load is shorter than that of the configuration 1, this leads to higher critical load and higher MOS_c which is extremely large. Thanks to this the beam ① in the last configuration could have a section smaller than $A_2 = 0.65 \text{ cm}^2$ and this would lead to a lighter structure.

3)NEW CONFIGURATION

Let's consider a new configuration sketched in fig.14 with the same **hypothesis** as the previous configurations:

- Material: generic aluminum with E_{Al} , σ_{yAl} and ρ_{Al} given in the initial data;
- Circular thin walled cross section with $I = 2.92 * 10^{-9} \text{ m}^4$;
- Cross section area $A_2 = 0.65 \text{ cm}^2$.

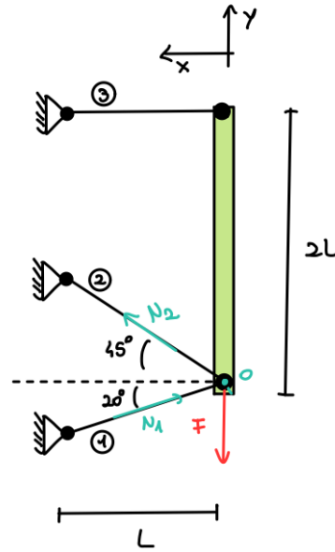


Figure 14: Configuration 4 with final load's distribution along each beam.

Computing each beam's length and equilibrium of forces and momentum:

$$l_3 = L = 30 \text{ cm}$$

$$l_1 * \cos 20^\circ = L \quad \text{so} \quad l_1 = \frac{L}{\cos 20^\circ} = 31.93 \text{ cm}$$

$$l_2 = \frac{L}{\cos 45^\circ} = 42.43 \text{ cm}$$

$$\begin{cases} \sum N_x = N_3 + N_2 \cos 45^\circ + N_1 \cos 20^\circ = 0 \\ \sum N_y = N_2 \sin 45^\circ - N_1 \sin 20^\circ - F = 0 \\ \sum M_o = -N_3 * 2L = 0 \end{cases}$$

So it is obtained that:

$N_3 = 0 \text{ N}$ so beam ③ is not loaded;

$N_1 = -\frac{F}{\cos 20^\circ * (1 + \tan 20^\circ)} = -4974.98 \text{ N}$ so beam ① is in compression;

$N_2 = \frac{N_1 \sin 20^\circ + F}{\sin 45^\circ} = 6611.39 \text{ N}$ so beam ② is in traction.

Assuming $N_{adm} = 16250 \text{ N}$ as the previous cases they can be evaluated the MOS for beam ① and ②:

$$\textcircled{2} \quad MOS_t = \frac{N_{adm}}{FOS * N_2} - 1 = 0.229 = 22.9\%$$

$$\textcircled{1} \quad MOS_c = \frac{P_{cr}}{FOS * N_1} - 1 = 0.987 = 98.7\% \quad \text{where} \quad P_{cr} = 19767.07 \text{ N}$$

Both MOS are admissible and the structural mass is:

$$M_{str} = \rho A_2 (L + l_1 + l_2) = 0.183 \text{ Kg}$$

Thanks to this new configuration it is possible to obtain a good *Margin Of Safety* for both the loaded beams and the best structural mass among all the considered configurations, in particular it is decreased by 17.93% respect to the configuration 3.

FINAL RESULTS

Configuration's type	1	2	3	4
MOS_t	80%	155%	14%	22.9%
MOS_c	24%	/	239%	98.7%
M_{str}	0.2234 Kg	/	0.223 Kg	0.183 Kg

Table 3: Final results of MOS and structural mass for all the considered configurations with circular thin walled cross section of area $A = 0.65 \text{ cm}^2$.